2-1

(a.)

X	y	$x-\bar{x}$	$(x-\bar{x})^2$	$(y-\bar{y})$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\Sigma X_i=5$	$\Sigma Y_i=10$	$\Sigma(x_i-\bar{x})=0$	$\Sigma(x_i-\bar{x})^2=10$	$\Sigma(y_i-\bar{y})=0$	$\Sigma(x_i-\bar{x})(y_i-\bar{y})=8$
$\bar{x}=1$	$\bar{y}=2$				

(b.)

$$y_i=B_1+B_2X_i+\varepsilon_i$$

$$\hat{y}_i=b_1+b_2X_i$$

$$b_2=\frac{\Sigma(x_i-\bar{x})(y_i-\bar{y})}{\Sigma(x_i-\bar{x})^2}=\frac{8}{10}=0.8$$

$$b_1=\bar{y}-b_2\cdot\bar{x}=2-0.8\cdot1=1.2$$

$$\Rightarrow \hat{y}_i=1.2+0.8x_i$$

b1為母體回歸線截距B1的估計值，其值為1.2，表示當x=0時，估計值ŷ=1.2
b2為母體回歸線斜率B2的估計值，其值為0.8，表示當x上升1單位時，估計值ŷ會上升0.8個單位

(c)

$$\sum_{i=1}^{5} X_i^2 = 3^2 + 2^2 + 1^2 + (-1)^2 + 0^2 = 15 \quad , \sum_{i=1}^{5} X_i Y_i = 3\times4 + 2\times2 + 1\times3 + (-1)\times1 + 0\times0 = 18$$

$$\begin{aligned}\Sigma(X_i-\bar{X})^2 = \Sigma(X_i^2 - 2X_i\cdot\bar{X} + \bar{X}^2) &= \Sigma X_i^2 - 2\cdot\bar{X}\cdot\Sigma X_i + N\cdot\bar{X}^2 \\ &= \Sigma X_i^2 - 2\cdot\bar{X}\cdot N\cdot\bar{X} + N\cdot\bar{X}^2 \\ &= \Sigma X_i^2 - 2N\bar{X}^2 + N\bar{X}^2 \\ &= \Sigma X_i^2 - N\bar{X}^2 = 15 - 5\times1^2 = 10 \ \#\end{aligned}$$

$$\begin{aligned}\Sigma(X_i-\bar{X})(Y_i-\bar{Y}) &= \Sigma[X_iY_i - X_i\bar{Y} - \bar{X}\cdot Y_i + \bar{X}\cdot\bar{Y}] \\ &= \Sigma X_iY_i - \bar{Y}\Sigma X_i - \bar{X}\cdot\Sigma Y_i + N\bar{X}\bar{Y} \\ &= \Sigma X_iY_i - \bar{Y}\cdot N\bar{X} - \bar{X}\cdot N\bar{Y} + N\bar{X}\bar{Y} \\ &= \Sigma X_iY_i - N\bar{X}\bar{Y} = 18 - 5\times1\times2 = 8 \ \#\end{aligned}$$

(d)

$$S_y^2 = \frac{\sum_{i=1}^{5}(Y_i-\bar{Y})^2}{5-1} = \frac{2^2+0^2+1^2+(-1)^2+(-2)^2}{4} = 2.5$$

$$S_x^2 = \frac{\sum_{i=1}^{5}(X_i-\bar{X})^2}{5-1} = \frac{4+1+0+4+1}{4} = 2.5$$

$$S_{xy} = \frac{\sum_{i=1}^{5}(X_i-\bar{X})(Y_i-\bar{Y})}{5-1} = \frac{4+0+0+2+2}{4} = 2$$

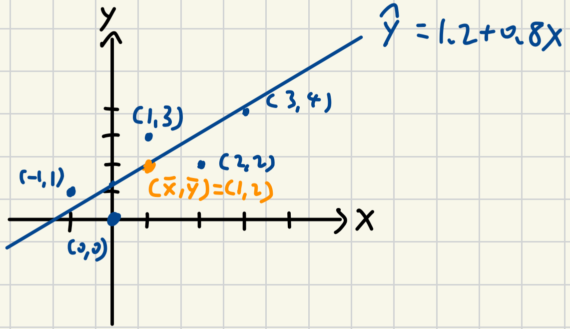
$$r_{xy} = \frac{S_{xy}}{S_x\cdot S_y} = \frac{2}{\sqrt{2.5}\cdot\sqrt{2.5}} = 0.8$$

$CV_x = 100\cdot\sqrt{2.5}/1 = 158.114$, $\hat{Y}_i = 1.2 + 0.8X_i$

median of $X = 1$, $\hat{e}_i = Y_i - \hat{Y}_i$

X_i	Y_i	$\hat{Y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$X_i\hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\Sigma X_i = 5$	$\Sigma Y_i = 10$	$\Sigma\hat{Y}_i = 10$	$\Sigma\hat{e}_i = 0$	$\Sigma\hat{e}_i^2 = 3.6$	$\Sigma X_i\hat{e}_i = 0$

(e.)



(f) 樣本迴歸線有通過 $(\bar{x},\bar{y})$ 之點 #

(g) $\bar{x}=1$, $\bar{y}=2$, $b_1=1.2$, $b_2=0.8$

$\bar{y}=b_1+b_2\bar{x} \ni 2=1.2+0.8\cdot 1$ #

(h) $\bar{\hat{y}}=\sum_{i=1}^{5}\hat{y}_i/5=\frac{3.6+2.8+2+0.4+1.2}{5}=2=\bar{y}$ #

(i) $\hat{\sigma}_1^2=\frac{\sum_{i=1}^{5}(y_i-b_1-b_2 x_i)^2}{5-2}=\frac{\sum_{i=1}^{5}\hat{e}_i^2}{3}=\frac{3.6}{3}=1.2$ #

(j) $\widehat{Var}(b_2|x)=\frac{\hat{\sigma}^2}{\sum(x_i-\bar{x})^2}=\frac{1.2}{10}=0.12$

$se(b_2)=\sqrt{\widehat{Var}(b_2|x)}=\sqrt{0.12}=0.3464$ #

2-22

(a)

```
Call:
lm(formula = totalscore ~ small, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max 
-191.63  -50.63   -7.12   42.37  324.38 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  918.628      2.536 362.260   <2e-16 ***
small          9.989      4.611   2.166   0.0305 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 73.36 on 1198 degrees of freedom
Multiple R-squared:  0.003903,	Adjusted R-squared:  0.003071 
F-statistic: 4.694 on 1 and 1198 DF,  p-value: 0.03047
```

b_1 (Intercept), b_2 small

$$TOTALSCORE_i = B_1 + B_2 SMALL_i + e_i$$

$$\widehat{TOTALSCORE_i} = b_1 + b_2 SMALL_i$$

$$\Rightarrow \widehat{TOTALSCORE_i} = 918.628 + 9.989 SMALL_i$$

b1=918.628代表普通班(SMALL=0)的期望平均總成績為918.628分
b2=9.989代表小班制的期望平均總成績比普通班高9.989分
在10%的顯著水準下，我們有90%的信心得出小班制有助於總成績提升的結論。

(b)

```
Call:
lm(formula = readscore ~ small, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max 
-76.123 -21.123  -3.356  15.877 192.877 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  434.123      1.044 416.021  < 2e-16 ***
small          5.466      1.897   2.881  0.00403 ** 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 30.19 on 1198 degrees of freedom
Multiple R-squared:  0.006882,	Adjusted R-squared:  0.006053 
F-statistic: 8.301 on 1 and 1198 DF,  p-value: 0.004032
```

$$\widehat{READSCORE_i} = 434.123 + 5.466 SMALL_i$$

可觀察出小班制的期望平均閱讀成績高於普通班的幅度小於總成績的幅度。

$9.989 > 5.466$

```
Call:
lm(formula = mathscore ~ small, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max 
-145.505  -35.028   -6.505   28.495  141.495 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  484.505      1.677 288.884   <2e-16 ***
small          4.522      3.049   1.483    0.138    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 48.52 on 1198 degrees of freedom
Multiple R-squared:  0.001832,	Adjusted R-squared:  0.0009992 
F-statistic: 2.199 on 1 and 1198 DF,  p-value: 0.1383
```

$$\widehat{MATHSCORE_i} = 484.505 + 4.522 SMALL_i$$

可觀察到斜率的估計值並無顯著，表示小班制的期望平均數學成績與普通班無顯著差異，其結論與總成績不同。

(c)

```
Call:
lm(formula = totalscore ~ aide, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max 
-195.14  -52.14   -6.95   40.95  330.86 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  922.145      2.640 349.250   <2e-16 ***
aide          -1.396      4.437  -0.315    0.753    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 73.5 on 1198 degrees of freedom
Multiple R-squared:  8.267e-05,	Adjusted R-squared:  -0.000752 
F-statistic: 0.09904 on 1 and 1198 DF,  p-value: 0.753
```

$\hat{\gamma}_1$ (Intercept), $\hat{\gamma}_2$ aide

$$TOTALSCORE_i = \gamma_1 + \gamma_2 AIDE_i + e_i$$

$$\widehat{TOTALSCORE} = 922.145 - 1.396 AIDE_i$$

$\hat{\gamma}1$=992.145代表沒有增加助教的普通班的期望平均總成績為992.145分
$\hat{\gamma}2$=-1.396代表增加助教的普通班期望平均總成績比沒有增加助教的普通班低1.396分
由於γ2並未顯準，故代表是否增加助教對於普通班的期望平均總成績並無顯著差異。

(d)

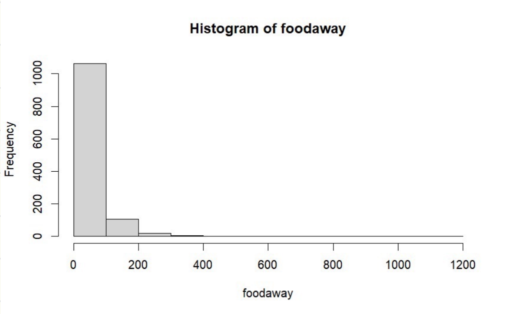
```
Call:
lm(formula = readscore ~ aide, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max 
-77.908 -21.536  -2.908  15.464 191.092 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 435.9084     1.0882 400.585   <2e-16 ***
aide         -0.3719     1.8285  -0.203    0.839    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 30.29 on 1198 degrees of freedom
Multiple R-squared:  3.453e-05,	Adjusted R-squared:  -0.0008002 
F-statistic: 0.04137 on 1 and 1198 DF,  p-value: 0.8389
```

```
Call:
lm(formula = mathscore ~ aide, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max 
-147.236  -32.236   -7.212   27.788  140.788 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  486.236      1.744 278.731   <2e-16 ***
aide          -1.024      2.931  -0.349    0.727    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 48.56 on 1198 degrees of freedom
Multiple R-squared:  0.0001019,	Adjusted R-squared:  -0.0007327 
F-statistic: 0.1221 on 1 and 1198 DF,  p-value: 0.7268
```

將應變數換成閱讀總成績以及數學總成績後，其截距的估計值依舊不顯著，表示是否增加助教均對普通班的期望平均閱讀成績以及期望數學成績並無顯著差異。

2-25

(a)




```
> summary(cex5$foodaway)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00   12.04   32.55   49.27   67.50 1179.00
```

mean : \$49.27 , median : \$32.55

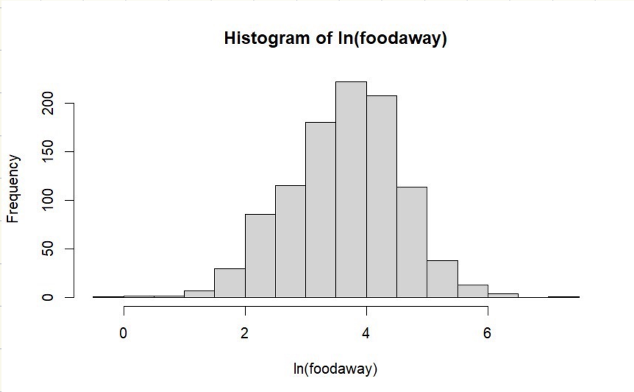
Q_1 = \$12.04 , Q_3 = \$67.50

(b)

```
> foodaway_adv<-cex5$foodaway[cex5$advanced==1]
> foodaway_col<-cex5$foodaway[cex5$college==1]
> foodaway_nei<-cex5$foodaway[cex5$college==0 & cex5$advanced==0]
> summary(foodaway_adv)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00   21.67   48.15   73.15   90.00 1179.00
> summary(foodaway_col)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00   14.44   36.11   48.60   68.67  416.11
> summary(foodaway_nei)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00    9.63   26.02   39.01   52.65  437.78
```

家庭裡有碩士學位的外食費用平均數為\$73.15，中位數為\$48.15
家庭裡有學士學位的外食費用平均數為\$48.60，中位數為\$36.11
家庭裡無碩士或學士學位的外食費用平均數為\$39.01，中位數為\$26.02

(c)




```
summary(ln_foodaway)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   -Inf   2.488   3.483    -Inf   4.212   7.072
```

從(b)小題可到foodaway的最小值為0，而ln(0)無意義，故取自然對數應將原本foodaway值為0的樣本剔除，所以ln(foodaway)的樣本數會比foodaway的樣本數還少。

(d)

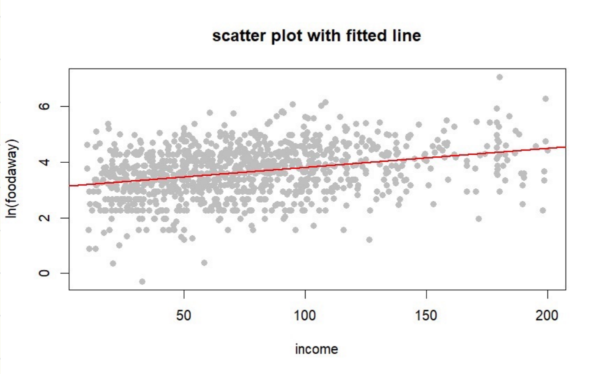
```
Call:
lm(formula = log(foodaway) ~ income, data = cex5_sub)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.6547 -0.5777  0.0530  0.5937  2.7000 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 3.1293004  0.0565503   55.34   <2e-16 ***
income      0.0069017  0.0006546   10.54   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8761 on 1020 degrees of freedom
Multiple R-squared:  0.09826,	Adjusted R-squared:  0.09738 
F-statistic: 111.1 on 1 and 1020 DF,  p-value: < 2.2e-16
```

$$\ln(FOODAWAY) = \beta_1 + \beta_2 INCOME + e$$

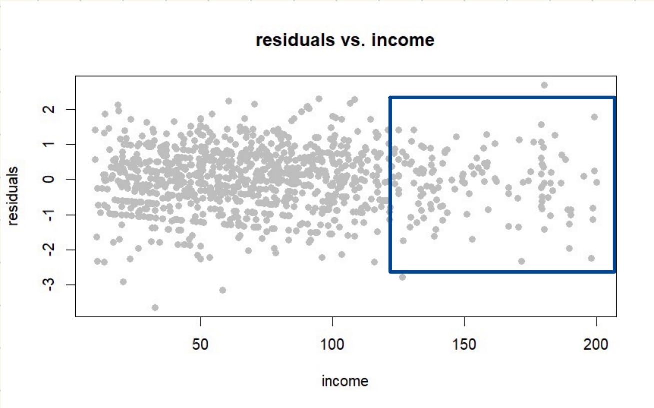
$$\Rightarrow \widehat{\ln(FOODAWAY)} = 3.1293004 + 0.0069017\, INCOME$$

斜率的估計值表示當家庭的月收入上升一單位(\$100)，
預期的外食支出會上升0.69017%。

(e)



(f)



從藍色框可觀察出隨著income增加，殘差點的擴散範圍有逐漸增大的傾向，代表在低收入的預測較準，高收入時預測誤差變大，可能存在異質變異的特徵。